

NAMWON, Korea, Republic of

WMO: 471730

Lat: 35.4214N

Lon: 127.3964E

Elev: 438

StdP: 14.46

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.4	15.0	-1.7	5.1	27.1	1.6	6.1	25.1	12.2	30.3	11.6	32.7	2.4	340	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.3	91.5	75.9	89.3	74.8	86.8	73.6	78.1	85.4	77.3	84.6	76.4	83.5	5.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.4	140.1	80.5	75.4	135.5	80.2	74.5	131.0	79.9	42.0	85.5	41.2	85.0	40.3	83.9	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.1	11.5	9.9	6.8	94.6	4.7	1.8	3.4	95.9	0.7	97.0	-1.9	98.0	-5.4	99.3		
			5.8	79.6	4.4	1.0	2.7	80.3	0.1	80.9	-2.4	81.5	-5.5	82.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.3	29.2	34.7	43.3	53.3	64.2	72.3	78.2	78.8	70.2	58.4	46.2	33.3
	DBStd	18.11	7.03	7.11	6.98	6.62	4.94	3.36	3.94	4.39	4.64	5.93	7.76	6.95
	HDD50	2043	644	431	233	44	0	0	0	0	0	8	164	519
	HDD65	4829	1109	849	674	352	73	1	0	0	9	216	563	983
	CDD50	3976	0	3	24	143	441	669	875	894	606	269	51	1
	CDD65	1288	0	0	0	1	50	220	410	429	165	12	1	0
	CDH74	10631	0	0	1	55	629	1474	3535	3925	965	45	2	0
	CDH80	3310	0	0	0	3	118	342	1180	1482	184	1	0	0
Wind	WSAvg	3.7	3.3	3.7	4.3	4.6	4.2	3.7	3.5	3.4	3.4	3.4	3.3	3.2
Precipitation	PrecAvg	52.0	1.2	1.7	2.0	4.0	3.7	7.2	12.2	8.9	5.6	2.3	1.8	1.3
	PrecMax	75.0	4.4	4.4	3.8	9.6	7.2	17.4	26.0	15.6	15.7	5.9	4.0	2.8
	PrecMin	31.5	0.3	0.1	0.3	0.7	0.1	0.5	1.6	2.9	0.5	0.2	0.6	0.0
	PrecStd	12.5	0.9	1.3	0.8	2.4	2.0	4.4	6.4	4.2	4.5	1.6	1.0	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.5	61.2	70.6	79.8	86.9	88.6	93.4	94.9	88.0	78.2	70.2	56.6
		MCWB	42.8	54.7	53.1	56.5	61.7	67.2	76.5	76.2	73.6	64.7	60.0	51.3
	2%	DB	47.1	55.7	65.7	75.1	83.1	86.0	91.0	92.5	84.4	74.8	65.3	51.3
		MCWB	39.4	45.9	50.3	55.2	60.8	67.5	76.2	76.1	71.1	62.2	56.9	45.0
	5%	DB	43.6	51.9	61.4	71.2	80.5	83.8	88.6	90.4	81.8	72.2	62.1	47.7
		MCWB	36.9	42.4	49.0	53.6	60.2	66.8	75.4	75.9	68.9	60.8	54.1	41.8
	10%	DB	40.2	47.5	57.5	67.6	77.6	81.7	86.3	87.9	79.6	69.6	58.7	44.4
		MCWB	34.4	39.6	47.4	52.1	60.1	66.6	74.9	75.6	67.9	59.3	52.2	39.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.6	56.7	57.9	61.8	70.1	74.6	79.2	79.6	76.5	70.2	63.5	53.7
		MCDB	49.1	59.8	61.8	69.6	73.6	80.1	85.6	87.6	83.8	74.1	68.6	54.7
	2%	WB	40.7	50.0	54.9	59.1	66.4	73.1	78.1	78.3	74.5	67.5	59.2	46.5
		MCDB	46.4	52.6	60.5	67.7	74.3	78.4	84.6	86.4	79.7	71.7	62.2	49.9
	5%	WB	38.1	43.9	51.9	57.2	63.9	71.1	77.3	77.5	72.9	64.5	56.2	42.7
		MCDB	42.4	49.4	60.3	64.3	73.5	77.3	84.4	85.1	78.1	68.0	60.3	46.5
	10%	WB	35.1	40.3	48.0	55.2	62.4	69.5	76.4	76.8	71.0	61.1	53.2	39.5
		MCDB	39.6	46.4	56.2	63.6	72.5	76.7	83.6	84.0	76.9	67.1	58.3	44.0
Mean Daily Temperature Range	5% DB	MDBR	18.1	18.9	22.0	23.7	23.4	17.2	12.9	14.3	17.0	21.0	19.5	16.6
		MCDBR	21.6	25.2	28.7	31.0	29.9	22.0	16.7	18.4	20.1	23.8	22.5	20.2
		MCWBR	13.9	15.6	15.4	13.5	10.5	6.4	4.7	4.8	7.5	10.9	12.8	13.9
	5% WB	MCDBR	18.6	20.3	21.8	19.3	20.0	13.6	13.9	14.8	14.1	12.6	17.1	17.6
		MCWBR	13.5	15.6	14.7	11.2	9.9	5.9	4.8	5.0	6.6	7.9	11.7	13.5
Clear-Sky Solar Irradiance	taub	0.438	0.482	0.563	0.587	0.600	0.664	0.616	0.570	0.498	0.471	0.452	0.435	
	taud	1.944	1.878	1.721	1.719	1.730	1.668	1.831	1.939	2.073	2.020	1.991	1.967	
	Ebn at Noon	227	233	225	229	228	214	223	231	241	234	221	218	
	Edh at Noon	46	55	70	74	74	78	66	58	49	47	43	42	
All-Sky Solar Radiation	RadAvg	813	1053	1382	1582	1743	1541	1357	1396	1275	1155	841	721	
	RadStd	57	84	127	126	175	143	190	193	137	103	93	47	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page